

Materials Engineering Reflection

Our goal for the materials engineering section was to build a sensor that could detect water vapor in the air. First, we learned what a sensor is and how they are applicable in life. Then, we headed to Wilbur's maker space lab to learn how to create a stencil of our IDE in the vinyl cutting machine and learn about the other equipment held there. We weeded our stencils and added transparent transfer tape to make the process of applying them to a microscope slide easier. Then we moved to Whitaker to learn about patterning of IDE's for detection. Here, we also applied our stencils to the slides and learned how to find the resistivity of copper by collecting its resistance, length, and cross-sectional area. At the beginning of our next class, we started in Sinclair to learn about MOCVDs and suited up to tour the clean room and to see photolithography and spincoating in action. For the second half of the class, we went back to Whitaker to measure the conductivity of silver paste, silver paint, and carbon paint. We were able to do this by creating masks on slides using tape and applying the desired material. We then scraped another slide across it so the material was evenly applied into the well made by the tape. After letting it dry and peeling off the tape, we were able to measure the length of the tape and calculate its cross-sectional area by gathering its width and the thickness of the tape. We found the resistance of the materials with the multimeter. With all of this data, we calculated the resistivity of the three materials.

The next class, we learned about the key components of a complete circuit and found that they rely on a path for electrons to flow in a complete loop. Since it has a direction of flow, it is polarized. We created a circuit schematic from our recently learned understanding of polarized and nonpolarized components. We constructed our butterfly circuits with LEDs (light-emitting diodes), batteries, copper tape, and a paper switch. I successfully created a working circuit through my understanding of the polarity of different materials and the direction of electron travel. Back at Whitaker, we learned that our detection material would be PEDOT-PSS because it's a semi-conducting polymer, so water interrupts its chains and changes its conductivity. We used spin-coating to apply the polymer to our IDEs. We finally tested our water vapor sensor by calculating its resistivity in ambient air and then breathing on it to add water vapor and calculating the new resistivity. Our sensor was effective because it measured a resistivity of 4,250 ohms per millimeter in ambient air and 450,000 ohms per millimeter when it detected water vapor.

In this section of Sensing the World, I learned about sensors and how to create one through the resources available to me on campus. I was exposed to collaborative hands-on learning that helped further my understanding of the materials past the threshold of lecture-style teaching. Even though finding the measurements needed for calculations was challenging, I was encouraged to ask questions that allowed me to figure it out and create a successful sensor with my group. I was able to improve my understanding of materials engineering and become comfortable with the information we explored.